

BABU BANARASI DAS UNIVERSITY



Predictive Analytics (PA)

(BCADSN15301)

LAB FILE

SUBMITTED TO:

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Section: BCADS32

Practical:5

Definition: You work for a telecommunication firm and have to combine a number of datasets into a single dataset for analyses and modelling later.

Purpose: To Show the Integration of telecommunications datasets using SPSS.

Learning: Importance of nodes such as Distinct, Append, Sort, Merge, Aggregate and Sample.

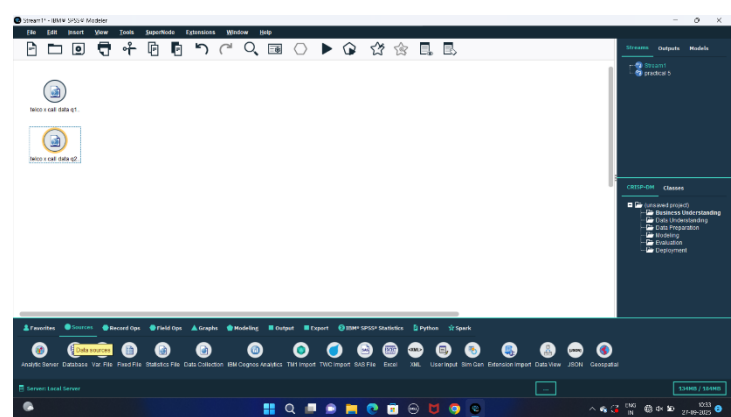
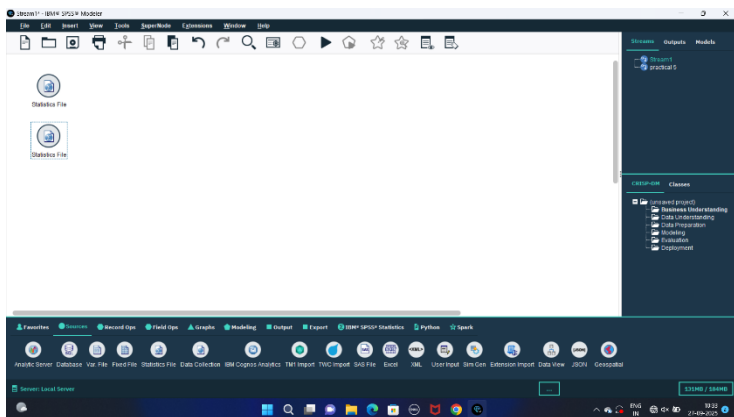
Required Tool: IBM SPSS tool

Working: Taking different files from Source and applying different Nodes.

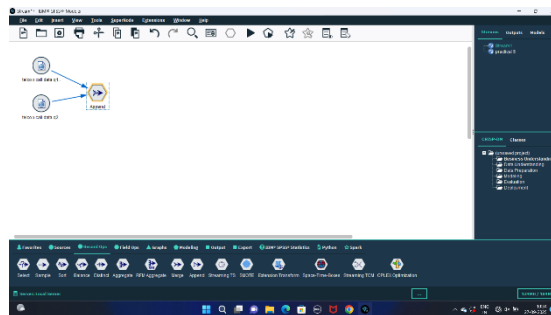
Step 1: Open SPSS Modeler in your System. For this practical, we will need 5 datasets which are as follows:

- telco x call data q1.sav
- telco x call data q2.sav
- telco x customer data.xlsx
- telco x products.dat
- telco x tariff.dat

We will begin by importing telco x call data q1.sav and telco x call data q2.sav files from Source Category. Both are statistics file so I'll choose Statistics Node.

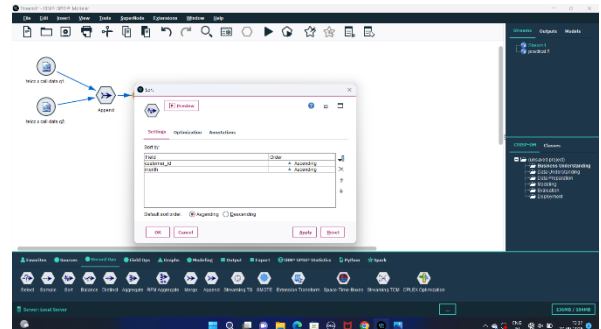
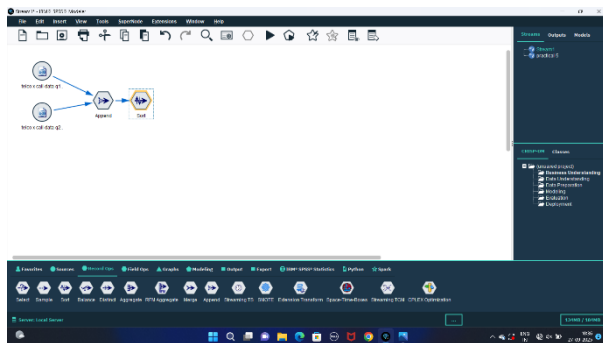


Step 2: From Record Ops I will choose Append Node and both my stats file will be connected to this Node. The Append Node will add the records of each file one after the other. Append Node works with Records of the Datasets.

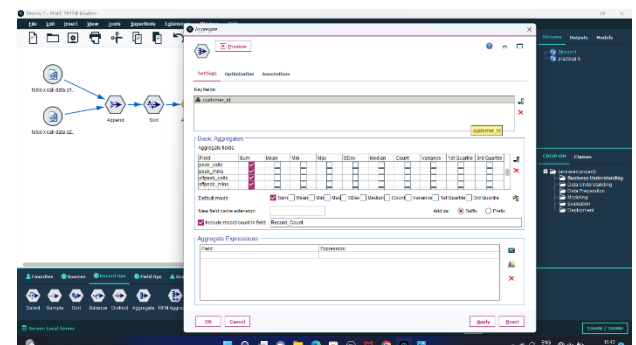
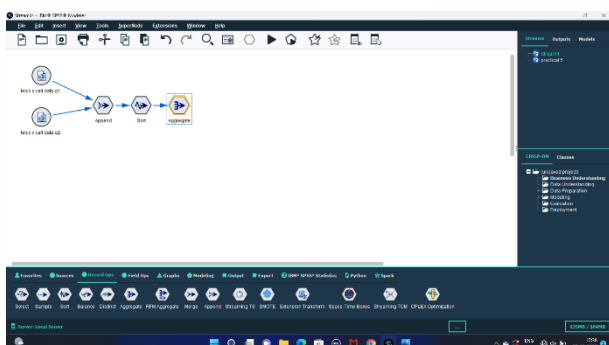


When you'll double click on the Append Node you'll notice all the fields from both the files.

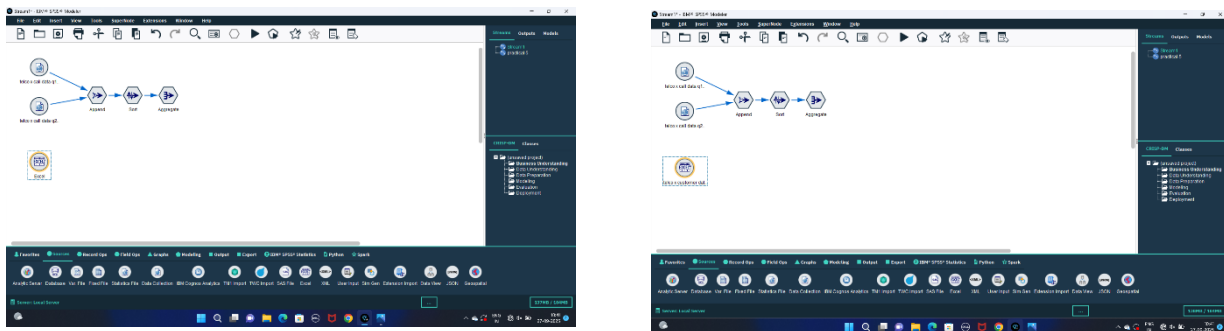
Step 3: From the Record Ops Category choose Sort Node. This node as the name suggests, sorts the dataset either in ascending or descending order. After connecting it with the Append node, double click on it and select the fields on the basis of which you want to sort. Here, I'm going with customer_id and month fields in ascending order.



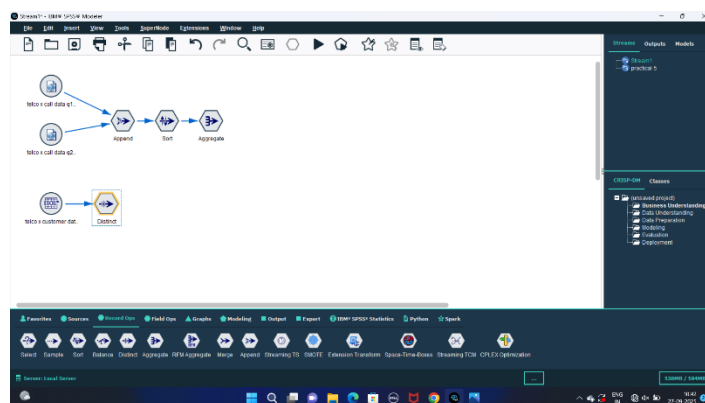
Step 4: From the Record Ops, choose Aggregate Node. This Node aggregates the value on the basis of Key(fields). This will aggregate the value of each Fields on the basis of one final Key. Key field here is customer_id and since, I want sum of all the fields I'll choose sum but can also chose mean, min, max, etc.



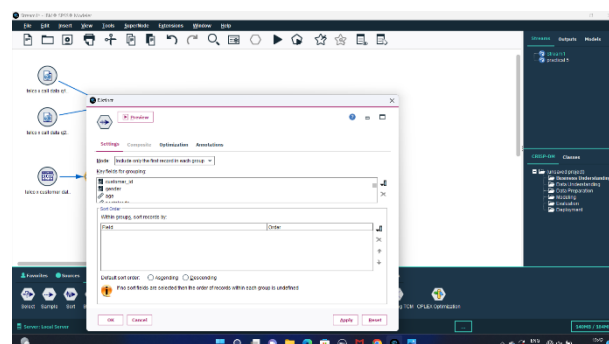
Step 5: Now, let's import our excel file named telco x customer data.xlsx from Sources Category.



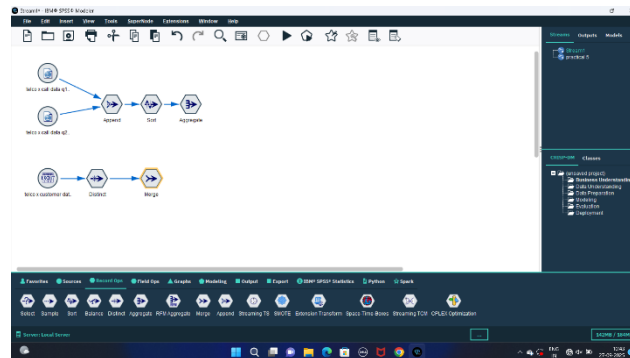
Step 6: From Record Ops, choose Distinct Node. This node will filter the duplicate records, a form of data-cleaning.



Step 7: Double click on the Distinct Node, and now in the Mode choose Include only the first record in each group. In key fields for grouping choose all the fields.

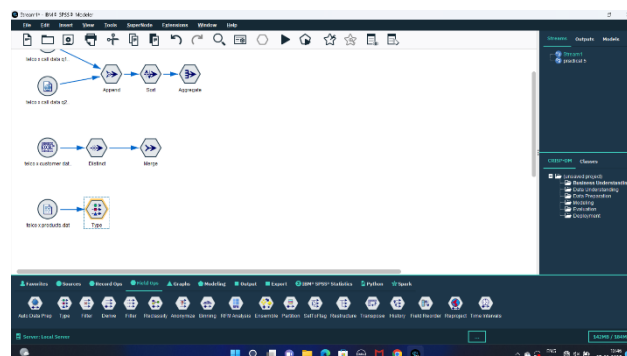


Step 8: From Record Ops, choose Merge Node. This node merges fields from two or more datasets. The Merge Node will share connections with Distinct node and from aggregate Node of previous two files.

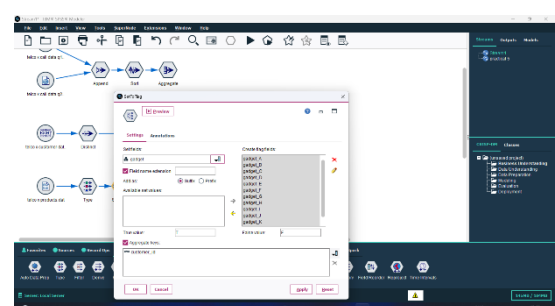
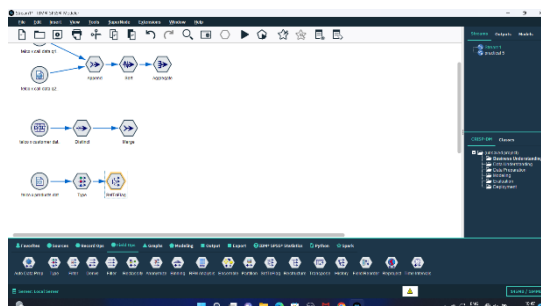


Step 9: Move back to Source Category and double click on var file. Now, import telco x products.dat file. Make sure to include Newline, Tabs.

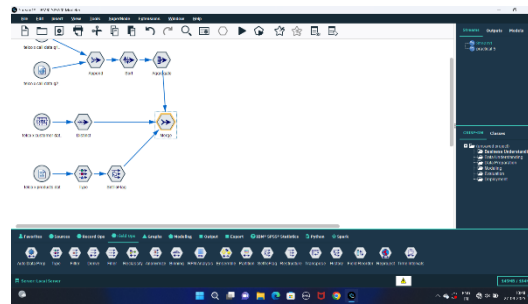
From Field Ops, choose Type Node. This node aids in understanding our fields measurement levels and storage.



Step 10: From the same category we will now pick settoflag Node. This node will help use to create new fields whose values will be of Flag type (ex: Y/N).



Step 11: Now make a connection with the already existing Merge Node. Merge node here, merges data from three sources.



Step 12: Connect a table from Output category and see how magnificently Merge Node works.

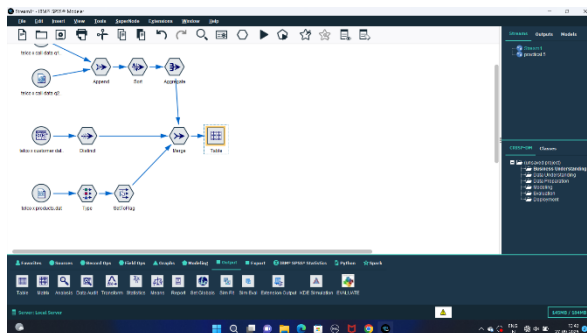
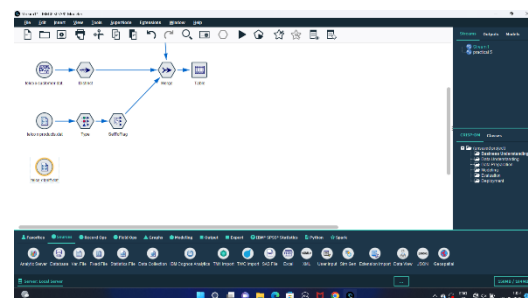
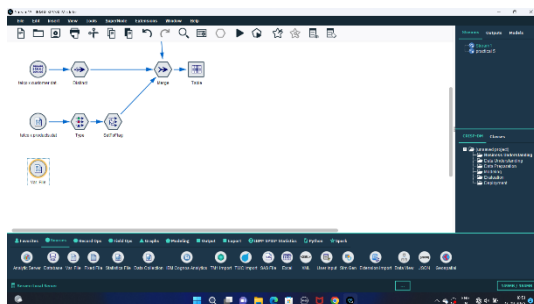
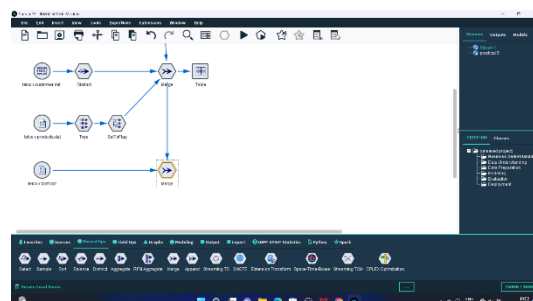


Table: Annotations.													
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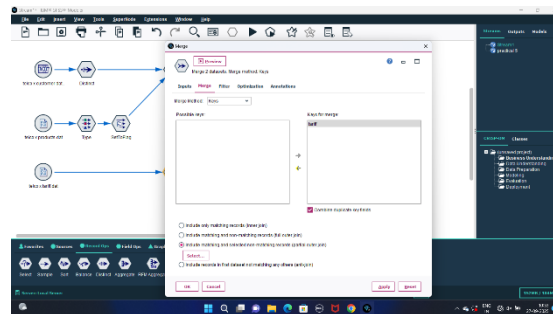
Step 13: Now let us import our last dataset which is again a var file. Import the file telco x tariff.dat



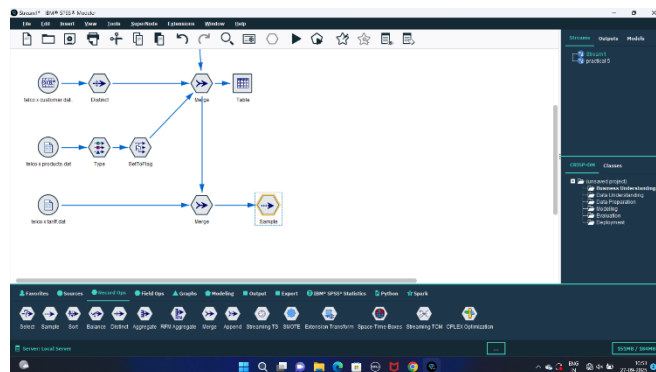
Step 14: Connect a Merge Node with this one. And the most interesting turn here will be that we will connect the previous Merge node with the New Merge Node.



Step 15: Double click on the New Merge Node. In the merge method select Keys. In possible keys opt for tariff and shift it to keys to merge section. Below it, choose for partial outer join.



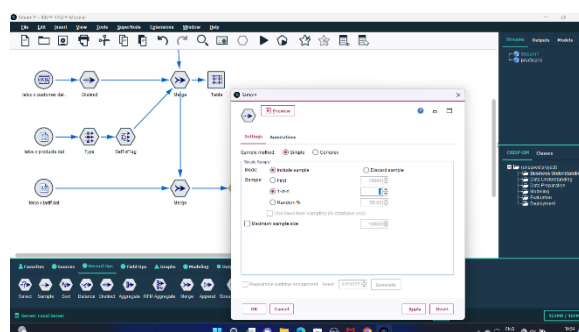
Step 16: From Record Ops, choose Sample Node. Now this node plays a key role. Because this will allow you to perform operations only on a handful of fields and you don't have to work on all the fields which can be cumbersome and unnecessary at times.



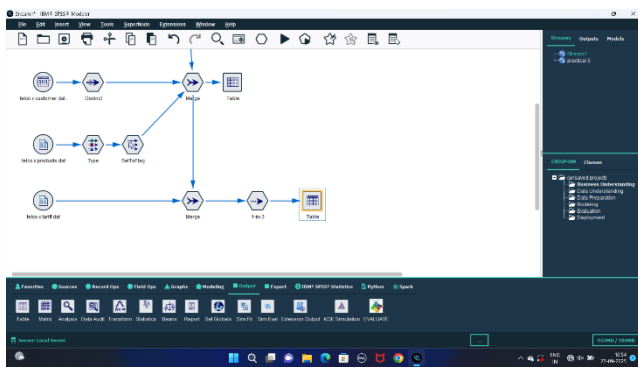
Step 17: When double clicked, it shows 3 methods in Sample, which are as follows:

- First: This allows you to select initial n records.
- 1 in n: This will provide you the records which are at nth position.
- Random %: When specified, it will provide you that % of records randomly.

Here, I am choosing 1 in n and I'm specifying n as 3. It means every 3rd record will be provided.



Step 18: Connect a Table to see the output.

[illegible]

This marks the end of this practical as well. Though a lengthy one but we got insights of some new Nodes as well which can actually be very helpful in time intensive task.

